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Microcontrollers Reflection

In this section of the module, we learned and experiment with microntrollers, programming, proper use of AI, prototyping, and documentation using GitHub. We were provided with our own Arduino ESP32 microcontrollers. We created our own GitHub repository to store our program files and document our work in README files. We worked through several projects using the Arduino and other electronics to solve problems. This taught us how to wire properly using breadboards, jumper wires, resistors, and other components. We learned the difference between power and ground, and how to use the power supplies and bench multimeters in the KeySight laboratory. After learning these fundamentals, we were assigned with creating our own project that would integrate our own sensor and Arduino with other electronics to solve a problem.

To be successful in this project we split up the responsibilities among our three team members, with one member doing coding, one doing the electronics wiring and setup, and the other building our physical elements. For this project, we were provided a wide range of materials. For electronics we had access to components such as potentiometers, push-button switches, servo, stepper, and DC motors, LEDs, and much more. Our physical materials included popsicle sticks, pipe cleaners, straws, Kinex, and more. The goal of our project was to detect a rise in humidity by wiring our sensor to ground and an Arduino digital PWM pin to detect differences in resistance. When the humidity rises, a vent will open using the servo motor, and a fan will blow that is attached to the DC motor.

To execute our design, we selected materials that were both applicable to our idea and easy to use. By dividing our responsibilities and working towards a well-defined end goal, we were very effective in completing the project. One challenge we encountered was interfacing the motors with the materials we had available. We first designed our fan out of popsicle sticks, but they weren't evenly distributed, which caused the fan to wobble when attached to the DC motor. Due to this, we changed the material to cardboard strips and measured the angles between the blades. By the end of our first lab period, we were able to complete all of the electronics and create working code, as well as finish our first physical prototype.